

Data sorting

Data sorting is any process that involves arranging the data into some meaningful order to make it easier to understand, analyze or visualize.

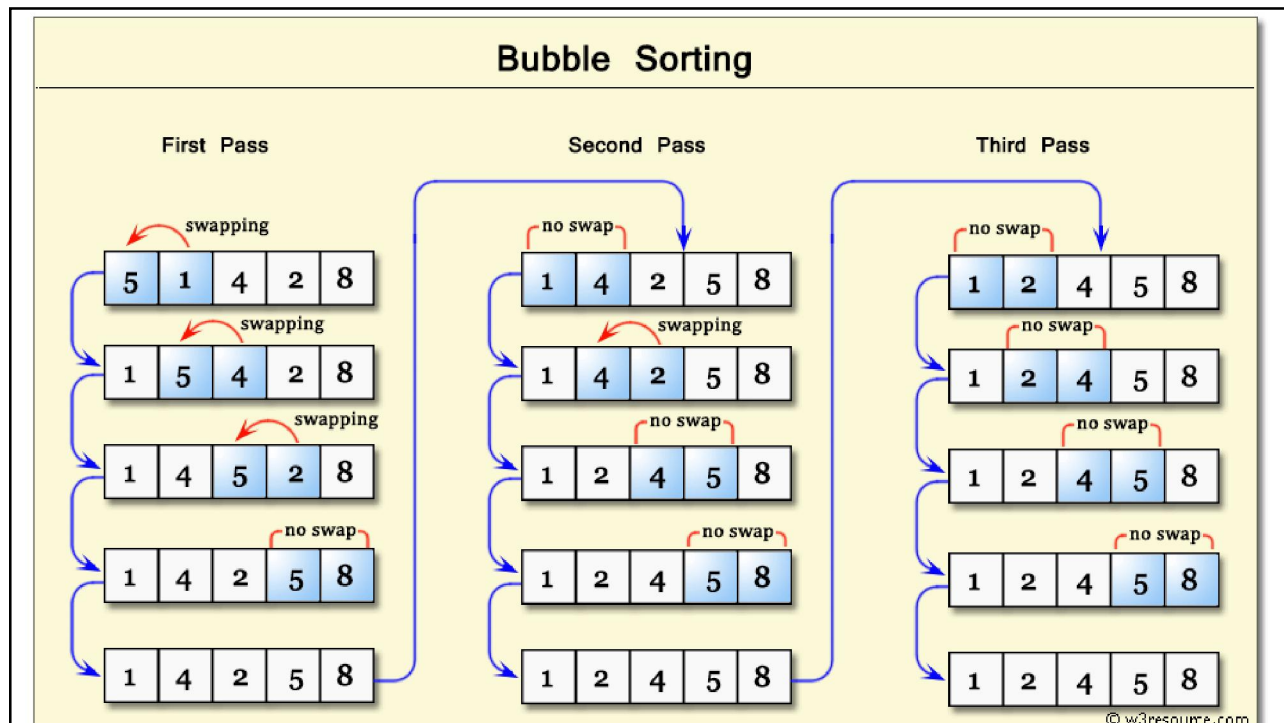
Bubble Sort

Bubble Sort is a simple algorithm which is used to sort a given set of **n** elements provided in form of an array with **n** number of elements. Bubble Sort compares all the element one by one and sort them based on their values.

Bubble Sort Algorithm

Following are the steps involved in bubble sort(for sorting a given array in ascending order):

1. Starting with the first element(index = 0), compare the current element with the next element of the array.
2. If the current element is greater than the next element of the array, swap them.
3. If the current element is less than the next element, move to the next element. **Repeat Step 1.**



```
for(i = 0; i < n; i++)  
{  
    for(j = 0; j < n-i-1; j++)  
    {  
        if( arr[j] > arr[j+1])  
        {  
            // swap the elements  
            temp = arr[j];  
            arr[j] = arr[j+1];  
            arr[j+1] = temp;  
        }  
    }  
}
```